Dongha Choi

School of Integrated Technology, Yonsei University
elijah1030@yonsei.ac.kr

SUMMARY

I am a Ph.D. candidate researching AI-RAN, semantic communications, and real-world wireless prototyping at the Intelligence Networking Lab., Yonsei University, Korea. My work spans learning-based communication theory, neural network interpretation, learned image compression, and over-the-air prototype validation. Before joining the lab, I earned a B.E. from the School of Integrated Technology at Yonsei University with early graduation. I served as a teaching assistant for three years and have three years of student council experience, including serving as president of the School of Integrated Technology and as a chief executive committee member of the College of Engineering. I also enjoy solving problems across social and medical domains through practical technological solutions.

EDUCATION

2023 - Present **Yonsei University**, Seoul, Republic of Korea
Ph.D. Candidate *(Advisor: Prof. Chan-Byoung Chae)*

2020 - 2022 **Yonsei University**, Seoul, Republic of Korea
B.E. in Integrated Technology *(GPA: 4.14/4.50, Early Graduation)*

PUBLICATIONS

Published Manuscripts

1. Hanju Yoo, **Dongha Choi**, Songkuk Kim, Chan-Byoung Chae, and Robert W. Heath, Jr., "Symbol Distributions in Semantic Communications: A Source-Channel Equilibrium Perspective," *IEEE Transactions on Communications*, vol. 74, pp. 13842-13855, 2026.
2. Hanju Yoo, **Dongha Choi**, Yonghwi Kim, Yoontae Kim, Songkuk Kim, Chan-Byoung Chae, and Robert W. Heath, Jr., "Bridging Neural Networks and Wireless Systems with MIMO-OFDM Semantic Communications," *IEEE Wireless Communications*, vol. 32, no. 5, pp. 48-55, 2025.
3. **Dongha Choi** and Chan-Byoung Chae, "A Study on the Transformer-based Equalizer in Semantic Communication Systems," in *Proceedings of the Symposium of the Korean Institute of Communications and Information Sciences*, pp. 962-963, 2024.

Manuscripts Under Review / In Preparation

1. **Dongha Choi**, Jungwoo Kim, Hyeonjin Lee, Yoonjung Lee, Jong-Seok Lee, and Chan-Byoung Chae, "Multimodal Semantic Communications for Physical AI: Shared Context Learning and Prototype Validation," *to be submitted*.
2. **Dongha Choi**, Hanju Yoo, Kyungpil Lee, Dongwook Kim, and Chan-Byoung Chae, "Explainable Semantic Communications: Neural Encoder Interpretation and Prototype Validation," under major revision for a special issue of *IEEE Communications Magazine*.
3. Hanju Yoo, **Dongha Choi**, Songkuk Kim, and Chan-Byoung Chae, "Back to the Bits: Parametric Symbol Compression for Semantic O-RAN Fronthaul," *to be submitted*.
4. **Dongha Choi**, Hanju Yoo, and Chan-Byoung Chae, "Steganography Semantic Communications: Architectures and Prototype Validation," *to be submitted*.

SYSTEM DEMONSTRATIONS

Dec. 8-12, 2025 **IEEE GLOBECOM 2025**, Taipei, Taiwan
Secure Semantic Communications with Steganography

Jun. 15-20, 2025 **IEEE IMS 2025**, San Francisco, USA
Real-Time Semantic Communications with Transparent AR Glasses (with Sensorview)

May 14-15, 2025 **IEEE 6G Summit Dresden 2025**, Dresden, Germany
Real-Time Semantic Communications with Transparent AR Glasses, as part of the Real-Time 6G Wireless Systems exhibition

AWARDS AND HONORS

Feb. 2020 **Best Dreamer Award**, high school valedictorian

Aug. 2021 **Highest Honors**, top 1% in the College of Engineering

Mar. 2022 **Full Scholarship for Undergraduate**
Issued by Kwanjeong Educational Foundation

Feb. 2025 **Honor Prize**, 31st Samsung Humantech Paper Award *(Co-recipient: Hanju Yoo)*

EXPERIENCES

Dec. 2024 - Jan. 2025 **Visiting Researcher**, Singapore University of Technology and Design
Hosted by Prof. Jihong Park; collaborative research on semantic communications.

2022 - 2025 **Teaching Assistant**, Yonsei University
Computational Thinking and SW Programming; Engineering Information Processing; Understanding the World with Data
Led weekly 4-hour Python, C, and SQL sessions and prepared course materials and examinations.

PROFESSIONAL SERVICE

Jul. 2026	**Reviewer**, IEEE Journal of Selected Topics in Signal Processing

SKILLS

Languages	Korean (Native), English (Fluent; TOEIC 920, TEPS 371), Chinese (Intermediate; HSK IV 280)
EECS	Communications Theory, Information Theory, Machine Learning (Deep Learning and Reinforcement Learning), Data Management, Digital System Design, and Computer Systems Programming
Programming	Python, TensorFlow, PyTorch, MATLAB, C (Fluent); JavaScript, HTML (Intermediate)